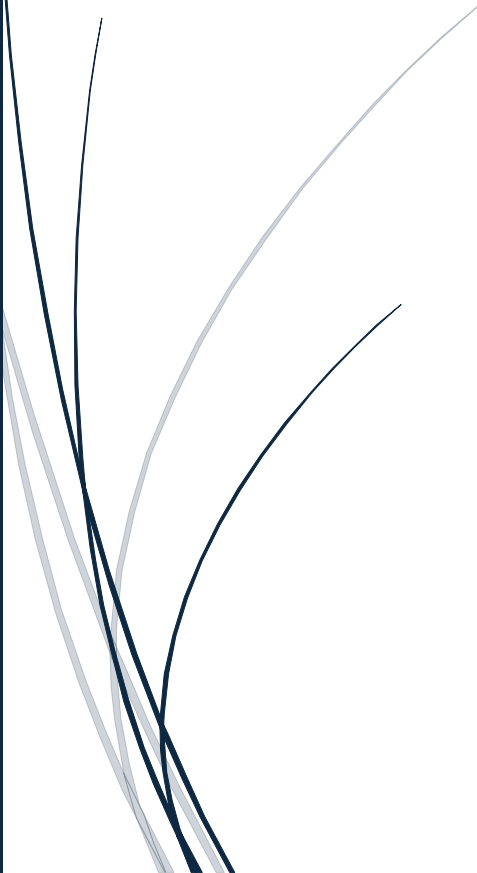




11/6/2025

Week-9 Lab:

Building & Testing a Secure VPC
Peering Connection on AWS



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Part 1: Environment Setup (In-Class Lab)

Both VPCs created correctly (distinct CIDRs), peering connection accepted, each route table contains exact peer-CIDR → pcx route; screenshot included

Step 1: Create LabVPC-A

Login to the AWS Console; select region us-east-1.

Search “VPC” → Create VPC.

Name: LabVPC-A; CIDR: 10.0.0.0/16; click Create.

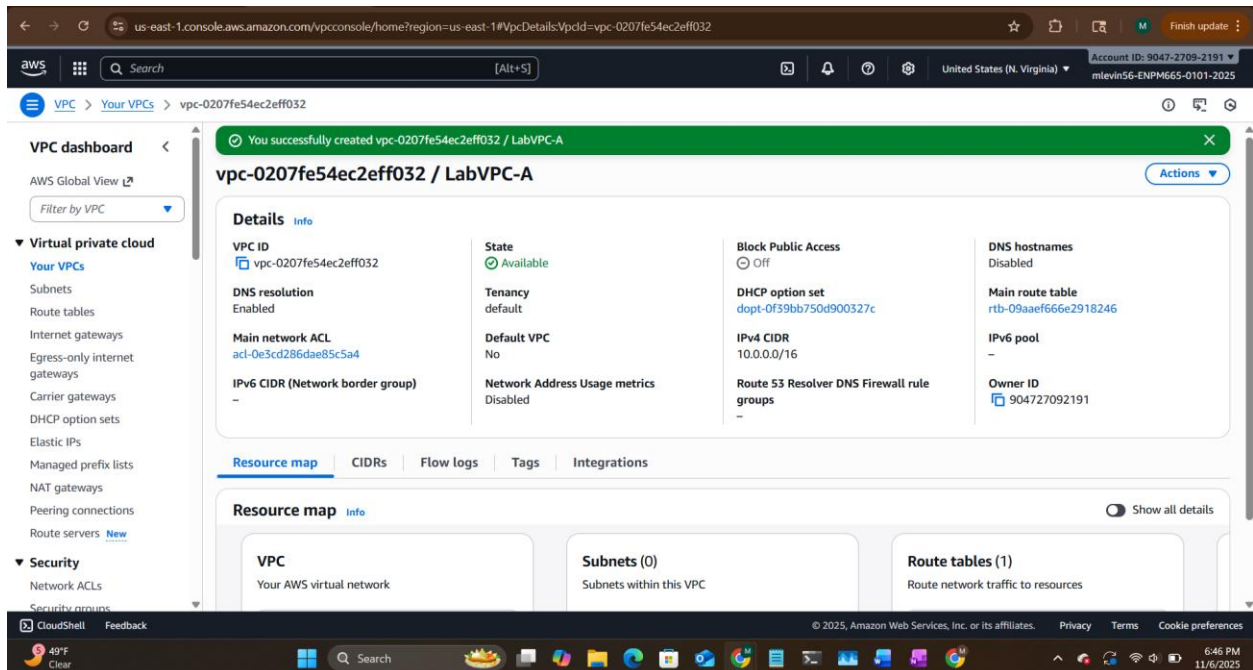


Figure 1 LabVPC-A creation

Step 2: Create & Configure LabA Public Subnet

VPC → Subnets → Create subnet → Name LabA-Subnet; CIDR 10.0.1.0/24; AZ us-east-1a → Create.

Select subnet → Actions → Modify auto-assign IPv4 → Enable → Save.

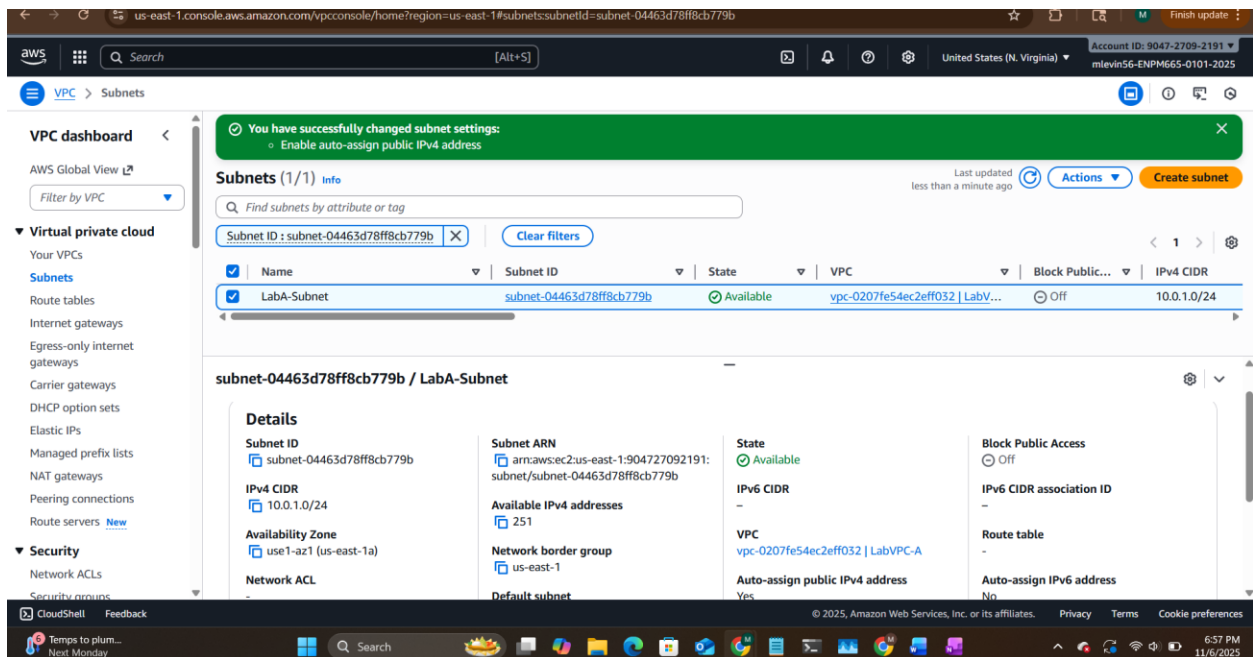


Figure 2 LabVPC-A Modify auto-assign IPv4: Enabled

Step 3: Attach Internet Gateway & Route Table for LabVPC-A

VPC → Internet Gateways → Create → Name LabA-IGW → Attach to LabVPC-A.

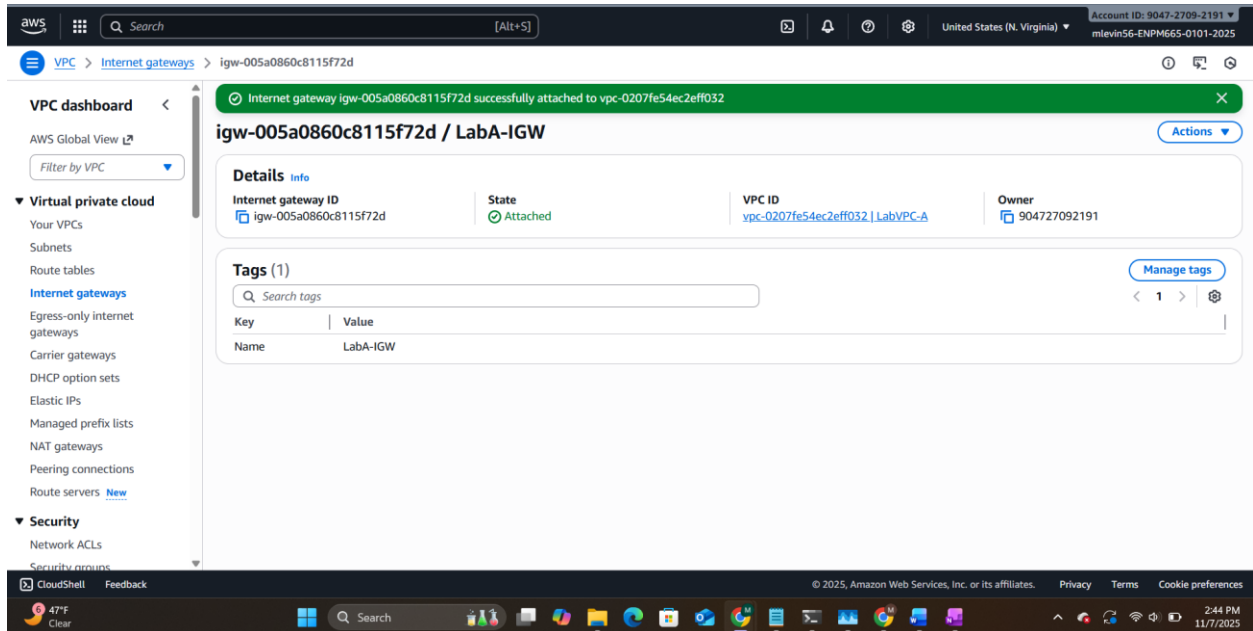


Figure 3 Attaching Internet Gateway Lab-IGW to LabVPC-A

VPC → Route Tables → Create route table → Name: LabA-PublicRT (VPC: LabVPC-A) → Edit routes → Add route: Destination 0.0.0.0/0 → Target LabA-IGW → Save

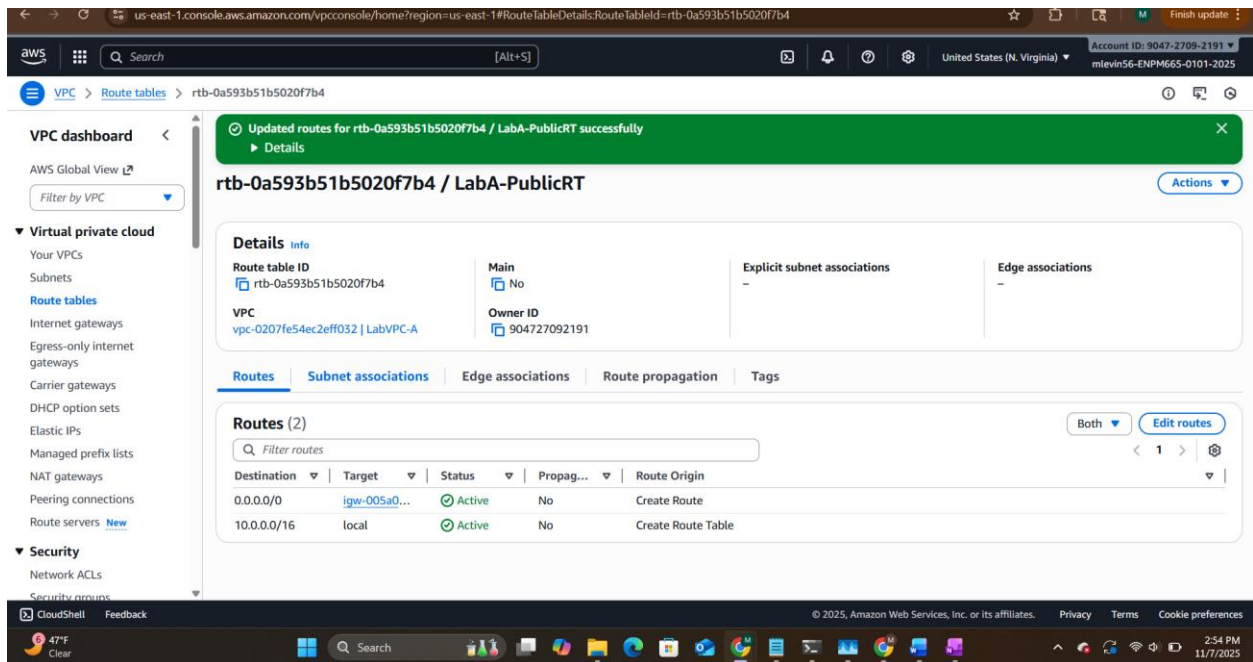


Figure 4 Adding the routes to LabA-IGW

Still in LabA-PublicRT, go to Subnet Associations → Edit subnet associations

Check LabA-Subnet → Save

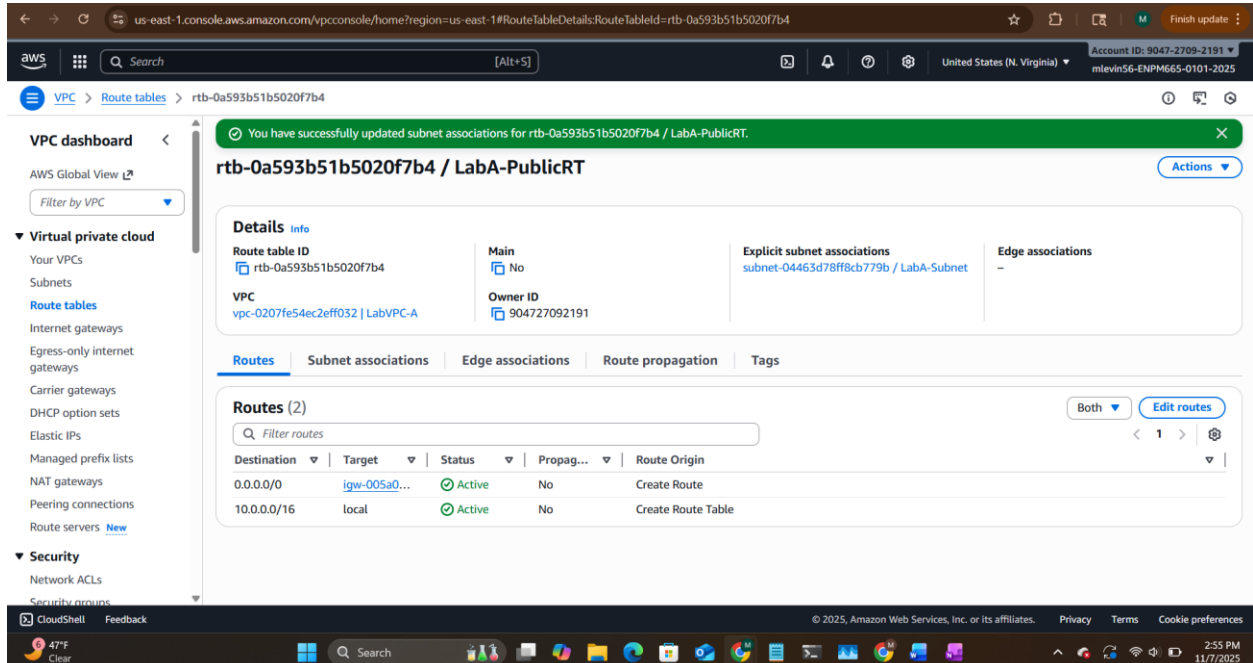


Figure 5 Subnet association

Step 4: Repeat for LabVPC-B

Create VPC named LabVPC-B with CIDR 10.1.0.0/16.

The screenshot shows the AWS Management Console interface for the 'Your VPCs' section. The left sidebar contains navigation links for VPC dashboard, AWS Global View, and Virtual private cloud. The main content area displays a table of VPCs with columns: Name, VPC ID, State, Block Public..., IPv4 CIDR, and IPv6 CIDR. The table lists three VPCs: LabVPC-A, LabVPC-B (selected), and LabVPC-C. Below the table, the details for 'vpc-0bf124bfc189154d4 / LabVPC-B' are shown, including VPC ID, State (Available), DNS resolution (Enabled), Main network ACL, IPv6 CIDR (Network border group), Tenancy (default), Default VPC (No), Network Address Usage metrics (Disabled), Block Public Access (Off), DHCP option set (dopt-0f39bb750d900327c), IPv4 CIDR (10.1.0.0/16), Route 53 Resolver DNS Firewall rule groups, DNS hostnames (Disabled), Main route table, IPv6 pool, and Owner ID (904727092191).

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR
LabVPC-A	vpc-02473e2cee2575d1d	Available	Off	172.31.0.0/16	-
LabVPC-B	vpc-0207fe54ec2eff032	Available	Off	10.0.0.0/16	-
LabVPC-C	vpc-0bf124bfc189154d4	Available	Off	10.1.0.0/16	-

Figure 6 LabVPC-B Creation

Create public subnet LabB-Subnet (CIDR 10.1.1.0/24), enable auto-assign IPv4.

The screenshot shows the AWS Management Console interface for the 'Subnets' section. A green notification banner at the top states: 'You have successfully changed subnet settings: Enable auto-assign public IPv4 address'. The left sidebar contains navigation links for VPC dashboard, AWS Global View, and Virtual private cloud. The main content area displays a table of subnets with columns: Name, Subnet ID, State, VPC, Block Public..., and IPv4 CIDR. The table lists four subnets: LabA-Subnet, LabB-Subnet (selected), LabC-Subnet, and LabD-Subnet. Below the table, the details for 'subnet-049731e2df084d4ca / LabB-Subnet' are shown, including Subnet ID, Subnet ARN, State (Available), IPv4 CIDR (10.1.1.0/24), Availability Zone (us-east-1), Network ACL (use1-az3 (us-east-1e)), Network border group (us-east-1), Default subnet (No), Auto-assign customer-owned IPv4 (Yes), Block Public Access (Off), IPv6 CIDR, VPC (vpc-0bf124bfc189154d4 | LabVPC-B), Auto-assign public IPv4 address (Yes), Route table, Auto-assign IPv6 address (No), and IPv6 CIDR reservations.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
LabA-Subnet	subnet-04463d78ff8cb779b	Available	vpc-0207fe54ec2eff032 LabV...	Off	10.0.1.0/24
LabB-Subnet	subnet-0e6ec6b894aa3eb40	Available	vpc-02473e2cee2575d1d	Off	172.31.0.0/20
LabC-Subnet	subnet-070503580d6286bba	Available	vpc-02473e2cee2575d1d	Off	172.31.32.0/2
LabD-Subnet	subnet-0e391078df5e02c4e	Available	vpc-02473e2cee2575d1d	Off	172.31.48.0/2

Figure 7 LabB-Subnet Creation

Create & attach IGW LabB-IGW, then update main route table: 0.0.0.0/0 → LabB-IGW.

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#RouteTableDetails:RouteTableId=rtb-0540c4865d29876e6

Updated routes for rtb-0540c4865d29876e6 / LabB-PublicRT successfully

rtb-0540c4865d29876e6 / LabB-PublicRT

Details

Route table ID: rtb-0540c4865d29876e6

VPC: vpc-0bf124bfc189154d4 | LabVPC-B

Main: No

Owner ID: 904727092191

Explicit subnet associations: subnet-049731e2df084d4ca / LabB-Subnet

Edge associations: -

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-037317a23956fc870	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table

Figure 8 Adding routes to LabB

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#InternetGateway:InternetGatewayId=igw-037317a23956fc870

Internet gateway igw-037317a23956fc870 successfully attached to vpc-0bf124bfc189154d4

igw-037317a23956fc870 / LabB-IGW

Details

Internet gateway ID: igw-037317a23956fc870

State: Attached

VPC ID: vpc-0bf124bfc189154d4 | LabVPC-B

Owner: 904727092191

Tags (1)

Key	Value
Name	LabB-IGW

Figure 9 Attaching the Internet gateway to LabB

Subnet associated with LabB-Subnet

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTableDetails:RouteTableId=rtb-0540c4865d29876e6

VPC dashboard < AWS Global View > Filter by VPC

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers New

Security
Network ACLs
Security groups

CloudShell Feedback

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You have successfully updated subnet associations for rtb-0540c4865d29876e6 / LabB-PublicRT.

rtb-0540c4865d29876e6 / LabB-PublicRT

Details info

Route table ID: rtb-0540c4865d29876e6

VPC: vpc-0bf124bfc189154d4 | LabVPC-B

Main: No

Owner ID: 904727092191

Explicit subnet associations: subnet-049731e2df084d4ca / LabB-Subnet

Edge associations: --

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-037317a23956fc870	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table

Figure 10 Association the subnet to LabB

Part 2: Launch EC2 Instances & Configure Security Groups

Step 5: Create Key Pair

EC2 → Key Pairs → Create Key Pair → Name LabKeyPair → Download .pem.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#KeyPairs:

EC2 > Key pairs

Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New

Images
AMIs
AMI Catalog

Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

Load Balancing
Load Balancers
Target Groups

CloudShell Feedback

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47°F Clear

3:14 PM 11/7/2025

Successfully created key pair

Key pairs (2)

Find Key Pair by attribute or tag

☐	Name	Type	Created	Fingerprint	ID
☐	LabKeyPair	rsa	2025/11/07 1...	3b:38:d1:71:90...	key-02f7f7a236c50b4ab
☐	MyLabKeyPair	rsa	2025/10/25 1...	22:af:98:47:dd...	key-082147a6f98b3f407

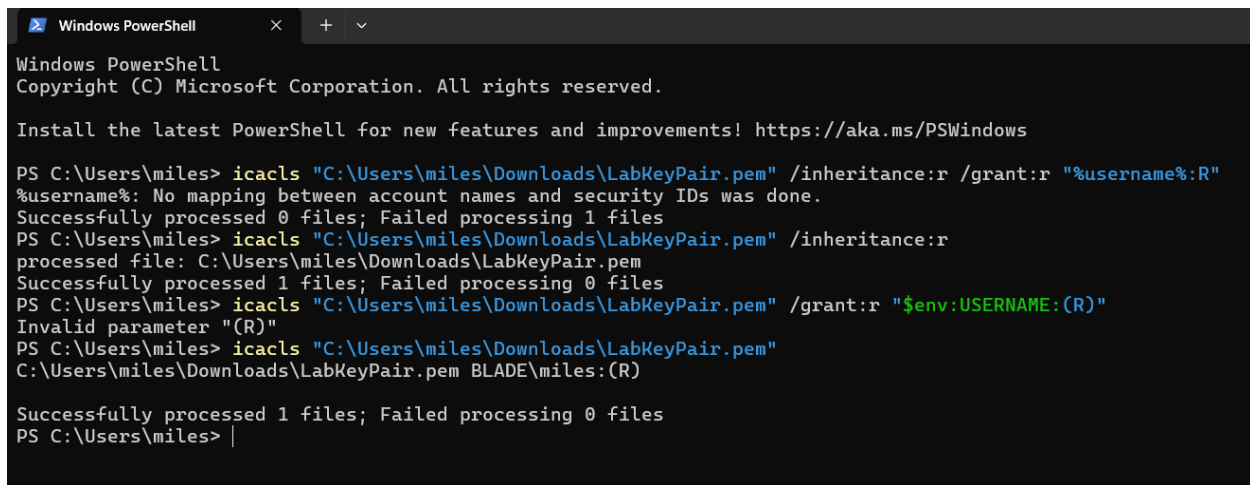
Recent download history

LabKeyPair.pem
1,678 B • Done

Full download history

Figure 11 Key Pair download

Modify the key permissions (Required for SSH access):



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\miles> icacls "C:\Users\miles\Downloads\LabKeyPair.pem" /inheritance:r /grant:r "%username%:R"
%username%: No mapping between account names and security IDs was done.
Successfully processed 0 files; Failed processing 1 files
PS C:\Users\miles> icacls "C:\Users\miles\Downloads\LabKeyPair.pem" /inheritance:r
processed file: C:\Users\miles\Downloads\LabKeyPair.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\Users\miles> icacls "C:\Users\miles\Downloads\LabKeyPair.pem" /grant:r "$env:USERNAME:(R)"
Invalid parameter "(R)"
PS C:\Users\miles> icacls "C:\Users\miles\Downloads\LabKeyPair.pem"
C:\Users\miles\Downloads\LabKeyPair.pem BLADE\miles:(R)

Successfully processed 1 files; Failed processing 0 files
PS C:\Users\miles> |
```

Figure 12 Modifying the key permissions

Security Groups restrict ICMP to peer CIDR only; no SSH open to 0.0.0.0/0; screenshot included

Step 6: Create Security Groups

For each of the EC2 instances, Create the below security groups

LabA-SG (Allow SSH and ICMP from VPC-B):

Inbound: Type=SSH, Source=MyIP

Inbound: Type=All ICMP-IPv4, Source=10.1.0.0/16

LabB-SG (Allow SSH and ICMP from VPC-A)

Inbound: Type=SSH, Source=MyIP

Inbound: Type=All ICMP-IPv4, Source=10.0.0.0/16

LabA-SG

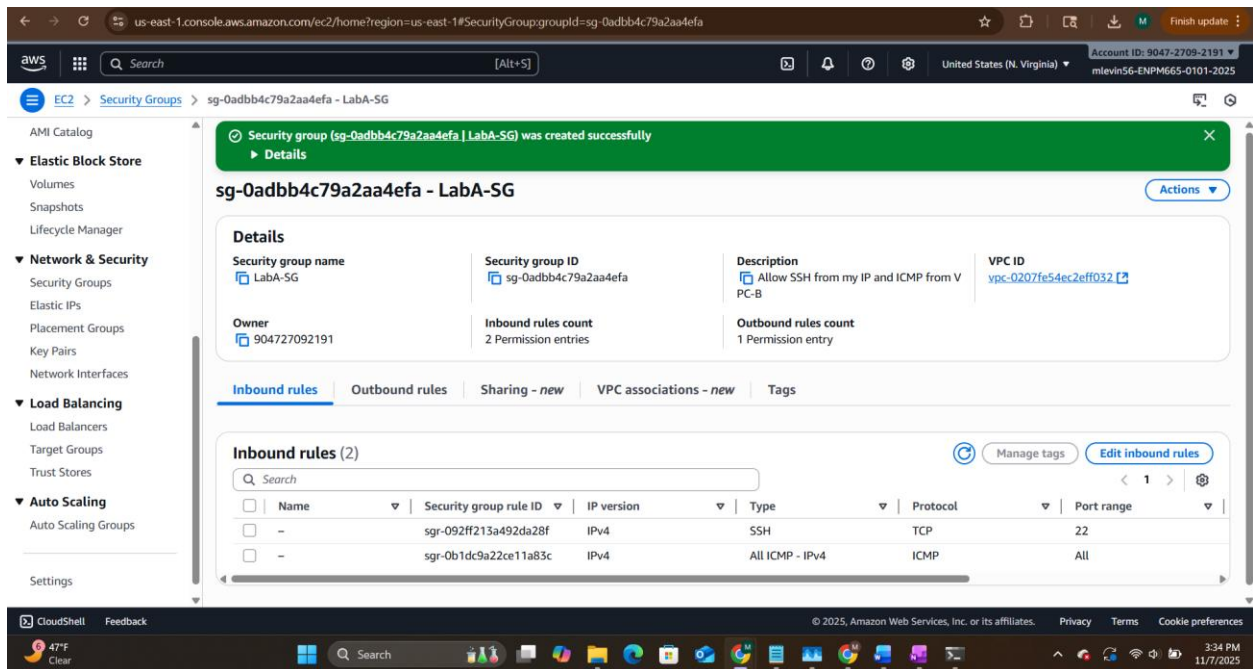


Figure 13 Adding the security group for LabA

LabB-SG

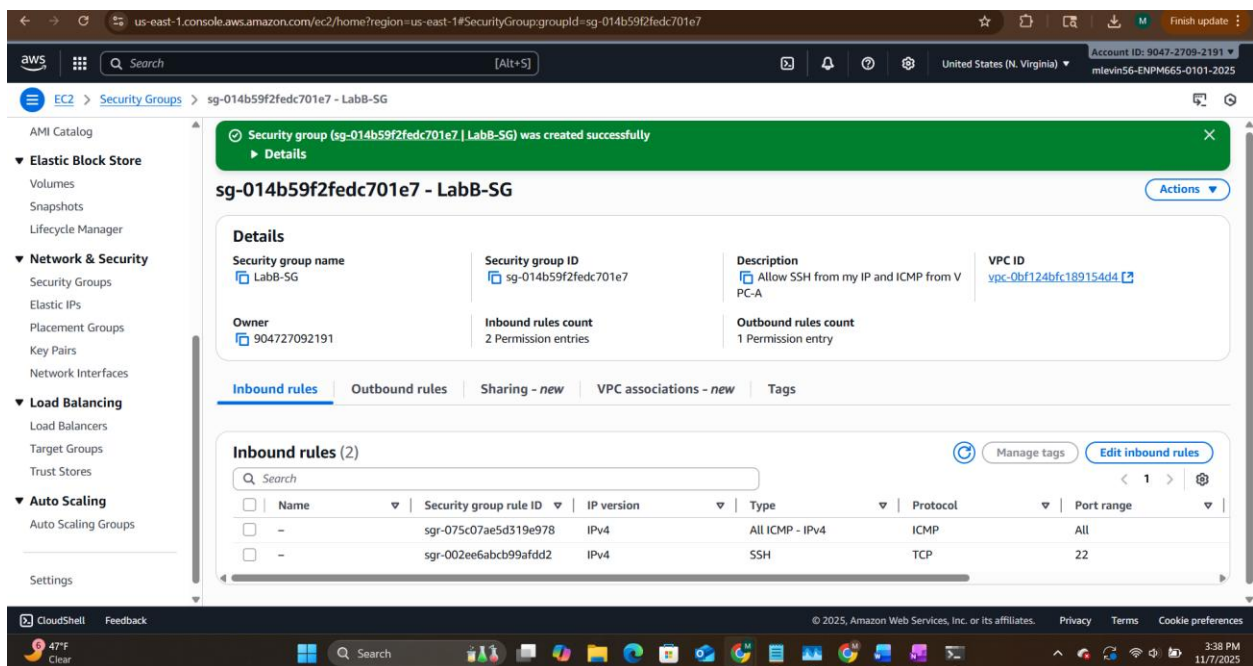


Figure 14 Adding the security group for LabB

Step 7: Launch EC2 Instances

EC2 → Launch Instance → Amazon Linux 2, t2.micro → Name LabA-Instance, Network=LabVPC-A, Subnet=LabA-Subnet, Auto-assign Public IP=Yes → Attach LabA-SG →

Key Pair LabKeyPair → Launch.

Network settings [Info](#)

VPC - required [Info](#)

vpc-0207fe54ec2eff032 (LabVPC-A)
10.0.0.0/16

Subnet [Info](#)

subnet-04463d78ff8cb779b LabA-Subnet
VPC: vpc-0207fe54ec2eff032 Owner: 904727092191 Availability Zone: us-east-1a (use1-az1)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

LabA-SG sg-0adbb4c79a2aa4efa X
VPC: vpc-0207fe54ec2eff032

Security groups that you add or remove here will be added to or removed from all your network interfaces.

[Compare security group rules](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 kernel-6.12 ...
ami-087b3cfeefdb81643

Virtual server type (instance type)
t3.micro

Firewall (security group)
LabA-SG

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Figure 15 Details for EC2 for LabA-Instance

Repeat for LabB: Name LabB-Instance, Network=LabVPC-B, Subnet=LabB-Subnet, SG=LabB-SG.

Network settings [Info](#)

VPC - required [Info](#)

vpc-0bf124bfc189154d4 (LabVPC-B)
10.1.0.0/16

Subnet [Info](#)

subnet-0ba477996c7585930 LabB-Subnet
VPC: vpc-0bf124bfc189154d4 Owner: 904727092191 Availability Zone: us-east-1a (use1-az1)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.1.1.0/24

Auto-assign public IP [Info](#)

Enable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

LabB-SG sg-014b59f2fedc701e7 X
VPC: vpc-0bf124bfc189154d4

Security groups that you add or remove here will be added to or removed from all your network interfaces.

[Compare security group rules](#)

Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 kernel-6.12 ...
ami-087b3cfeefdb81643

Virtual server type (instance type)
t3.micro

Firewall (security group)
LabB-SG

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Figure 16 Details for EC2 for LabB-Instance

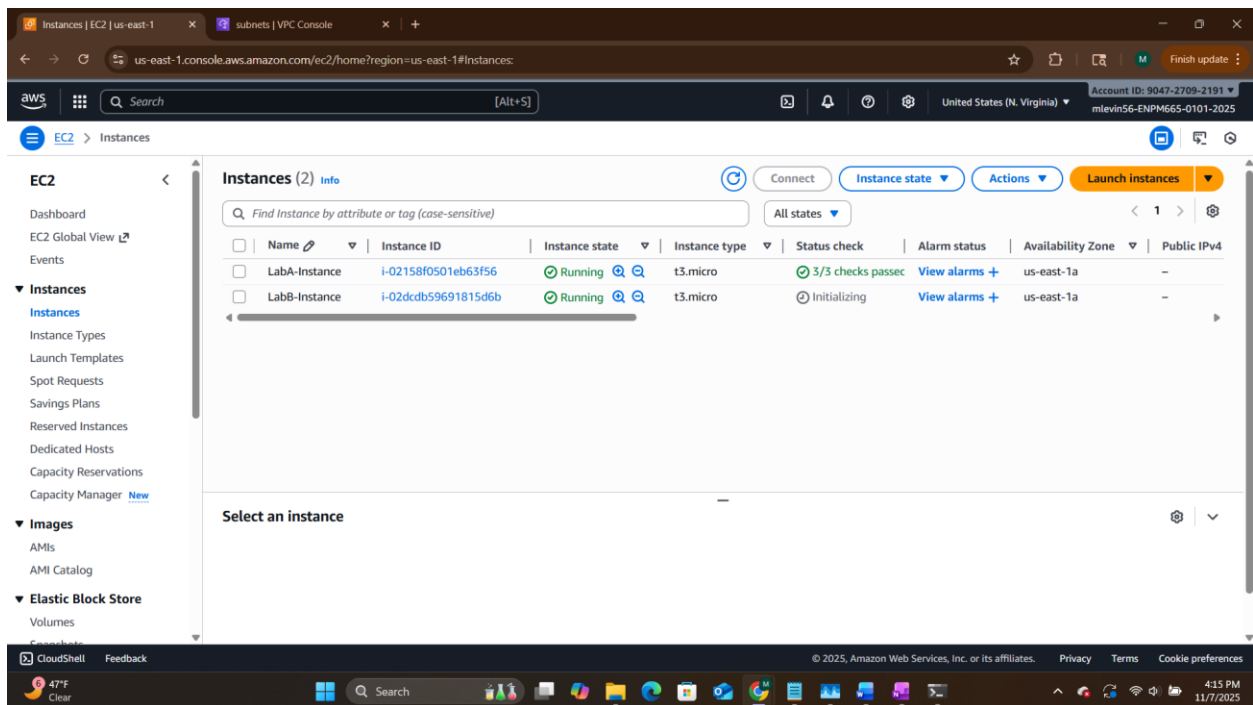


Figure 17 Launching both instances

Part 3: VPC Peering Configuration & Testing

Step 8: Create & Accept VPC Peering

VPC → Peering Connections → Create → Name LabVPC-Peering; Requester=LabVPC-A; Acceptor=LabVPC-B → Create → Accept.

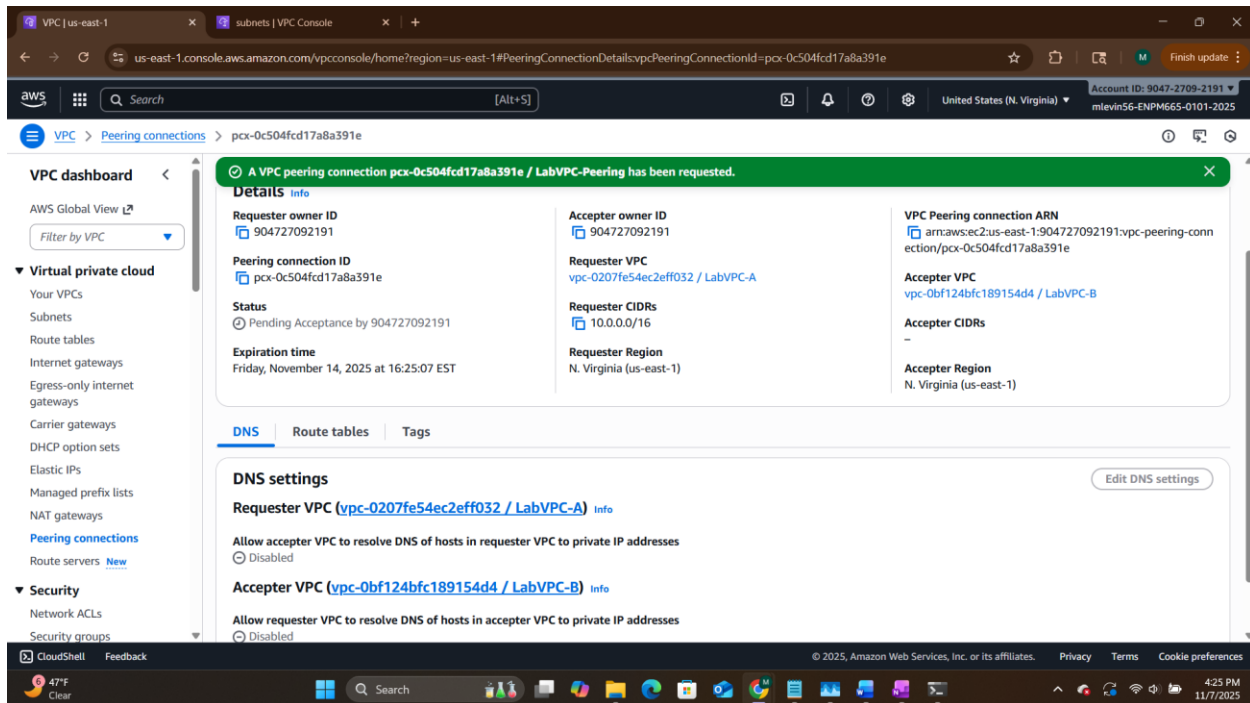


Figure 18 VPC peering

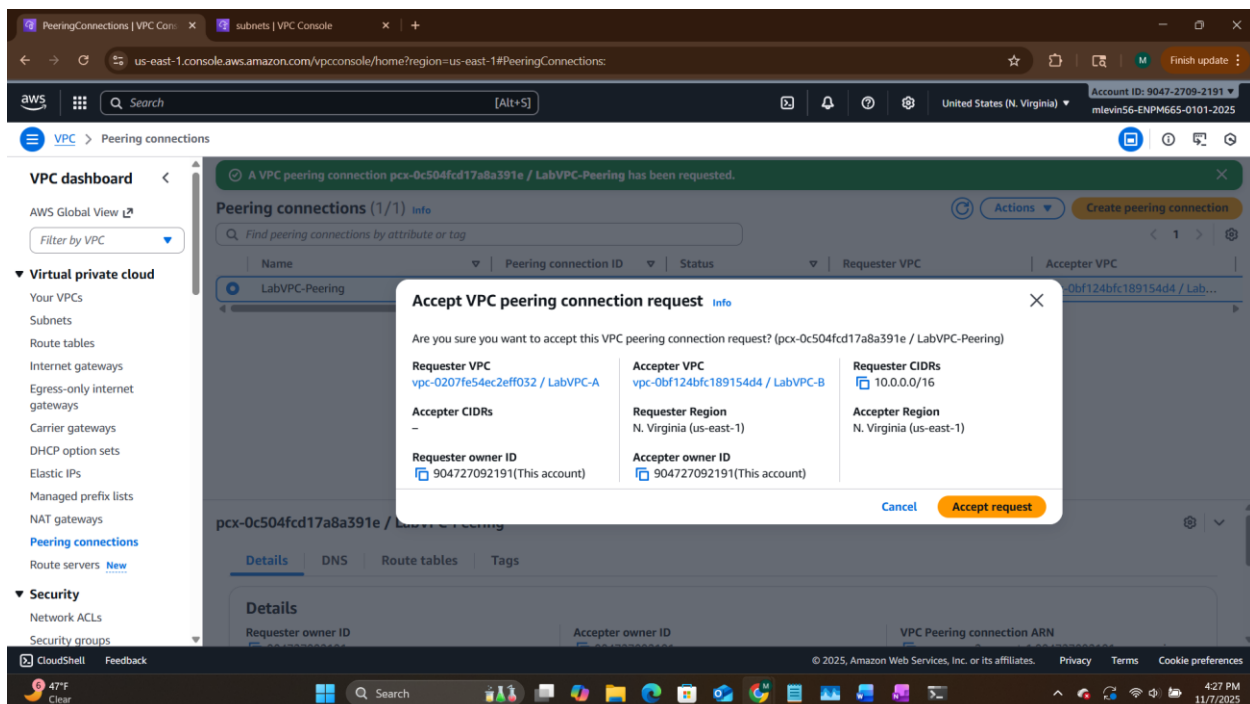


Figure 19 Accepting the peering connection request

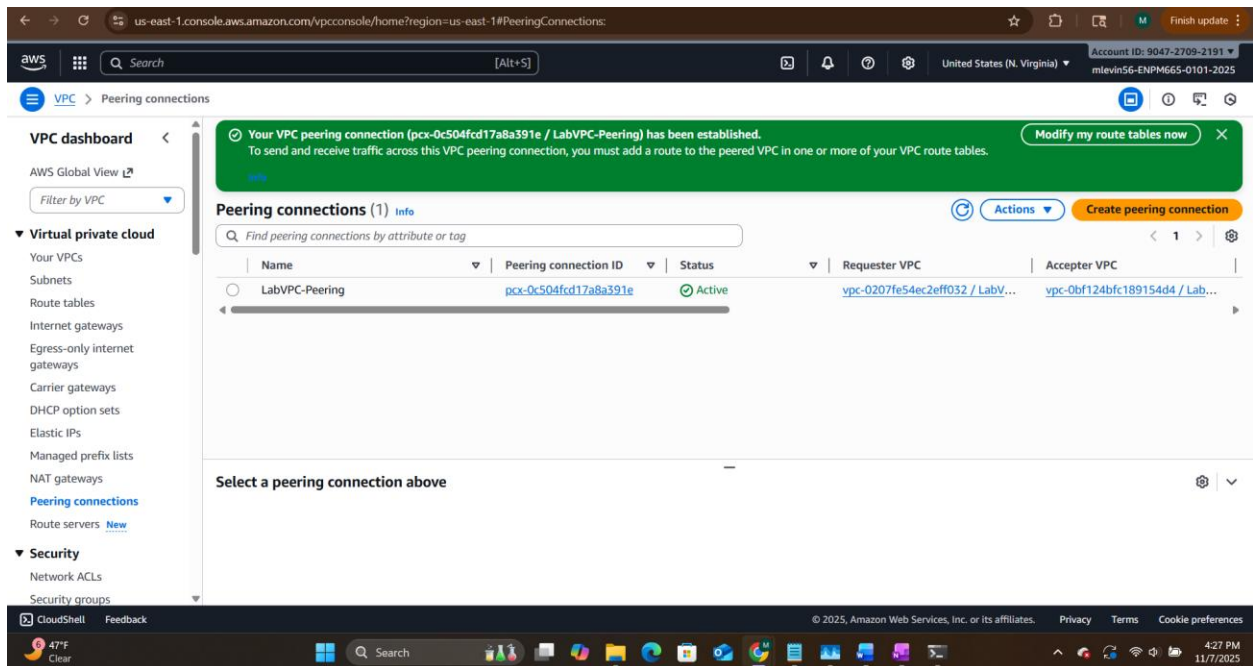


Figure 20 Establishing the VPC peering connection

Step 9: Update Route Tables for Peering

LabA-PublicRT: Destination=10.1.0.0/16 → Target=pcx-xxxxxx

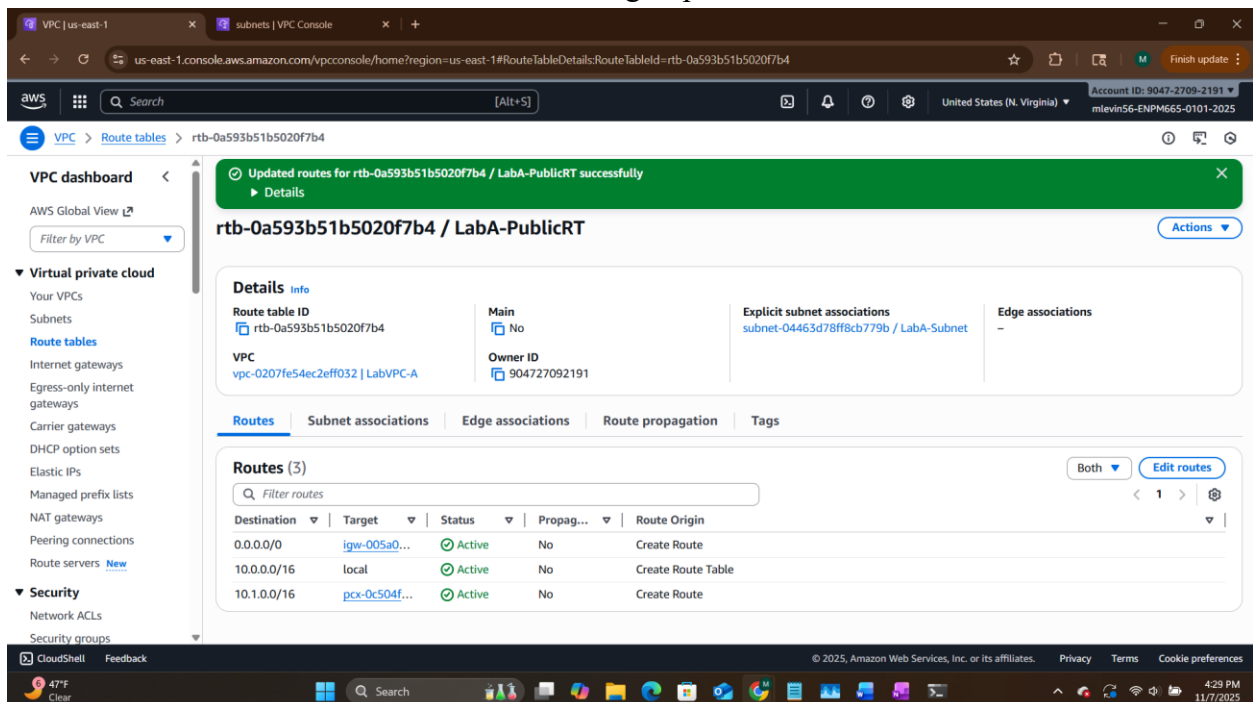


Figure 21 Updating route table for LabA

LabB-PublicRT: Destination=10.0.0.0/16 → Target=pcx-xxxxxx

VPC dashboard < AWS Global View Filter by VPC

Virtual private cloud Your VPCs Subnets Route tables Internet gateways Egress-only internet gateways Carrier gateways DHCP option sets Elastic IPs Managed prefix lists NAT gateways Peering connections Route servers New

Security Network ACLs Security groups

Updated routes for rtb-0540c4865d29876e6 / LabB-PublicRT successfully Details

rtb-0540c4865d29876e6 / LabB-PublicRT Actions

Details info

Route table ID rtb-0540c4865d29876e6 Main No Explicit subnet associations - Edge associations -

VPC vpc-0bf124bfc189154d4 | LabVPC-B Owner ID 904727092191

Routes Subnet associations Edge associations Route propagation Tags

Routes (3) Filter routes

Destination	Target	Status	Propag...	Route Origin
0.0.0.0/0	igw-03731...	Active	No	Create Route
10.0.0.0/16	pcx-0c504f...	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table

Both Edit routes

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Figure 22 Updating route table for LabB

Step 10: Verify Connectivity

SSH into LabA: `ssh -i LabKeyPair.pem ec2-user@<LabA_Public_IP>`.

```
ec2-user@ip-10-0-1-150 ~$ ssh -i "C:\Users\miles\Downloads\LabKeyPair.pem" ec2-user@44.197.189.187
The authenticity of host '44.197.189.187 (44.197.189.187)' can't be established.
ED25519 key fingerprint is SHA256:RhSa85VwOs1ljzLaS90TXe2xQ2ShoDLzuVDVUvA7V4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.197.189.187' (ED25519) to the list of known hosts.
```

```
#  
#####  
      \#####  
     /#####  
    /#####  
   /#####  
  /#####  
 /#####  
/#####  
#####
```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
[ec2-user@ip-10-0-1-150 ~]$ ping -c 4 10.1.1.xxx  
ping: 10.1.1.xxx: Name or service not known  
[ec2-user@ip-10-0-1-150 ~]$ ping -c 4 10.1.1.160  
PING 10.1.1.160 (10.1.1.160) 56(84) bytes of data.  
  
--- 10.1.1.160 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3090ms  
  
[ec2-user@ip-10-0-1-150 ~]$ ping -c 4 10.1.1.160  
PING 10.1.1.160 (10.1.1.160) 56(84) bytes of data.  
  
--- 10.1.1.160 ping statistics ---  
4 packets transmitted, 0 received, 100% packet loss, time 3135ms  
  
[ec2-user@ip-10-0-1-150 ~]$ ping -c 4 10.1.1.160  
PING 10.1.1.160 (10.1.1.160) 56(84) bytes of data.  
64 bytes from 10.1.1.160: icmp_seq=1 ttl=127 time=0.194 ms  
64 bytes from 10.1.1.160: icmp_seq=2 ttl=127 time=0.220 ms  
64 bytes from 10.1.1.160: icmp_seq=3 ttl=127 time=0.203 ms  
64 bytes from 10.1.1.160: icmp_seq=4 ttl=127 time=0.212 ms  
  
--- 10.1.1.160 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3120ms  
rtt min/avg/max/mdev = 0.194/0.207/0.220/0.009 ms  
[ec2-user@ip-10-0-1-150 ~]$ |
```

Repeat from LabB to LabA.

[illegible]

Figure 24 Ping testing to LabA from ssh LabB

Successful bidirectional ping (LabA→LabB and LabB→LabA); screenshot included

Lab completed without errors (“No troubleshooting needed”) OR documented clear debugging steps (screenshots + explanation) if issues arose

Troubleshooting & Problem-Solving

At first, all pings from LabA to LabB failed with 100% packet loss. The issue was that LabB-Subnet was not associated with the correct route table. Without that route, LabB did not know how to reach LabA’s network. After associating LabB-Subnet with LabB-PublicRT, private connectivity worked instantly. Both directions showed 0% packet loss, confirming successful routing and security group alignment.

Documentation & Reflection

This lab showed how correct subnet-to-route table associations are critical for VPC communication. You learn that even small routing errors stop connectivity, so always verify associations before testing network links.